

Collapsed Likelihood and VEB Algorithm for RSS Fine-Mapping with LD Uncertainty

1 Original model

Let $v \in \mathbb{R}^p$ denote the vector of summary statistics, for example

$$v = \frac{1}{N} X^\top y,$$

and let $\bar{R}$ be an out-of-sample reference LD matrix.

We consider the model

$$R \sim \text{IW}_p(\nu_0 \bar{R}, \nu_0 + p + 1), \tag{1}$$

$$v \mid R, b \sim \mathcal{N}_p(Rb, R/N). \tag{2}$$

The inverse-Wishart parameterization is chosen so that

$$\mathbb{E}(R) = \bar{R}.$$

Indeed, if

$$R \sim \text{IW}_p(S, m),$$

then

$$\mathbb{E}(R) = \frac{S}{m - p - 1},$$

and here

$$S = \nu_0 \bar{R}, \quad m = \nu_0 + p + 1.$$

The likelihood is

$$p(v \mid R, b) = \left(\frac{N}{2\pi} \right)^{p/2} |R|^{-1/2} \exp \left\{ -\frac{N}{2} (v - Rb)^\top R^{-1} (v - Rb) \right\}.$$

The quadratic term expands as

$$(v - Rb)^\top R^{-1} (v - Rb) = v^\top R^{-1} v - 2b^\top v + b^\top Rb. \tag{3}$$

Therefore,

$$p(v \mid R, b) = p(v \mid R, b = 0) \exp \left\{ Nb^\top v - \frac{N}{2} b^\top Rb \right\}. \tag{4}$$

2 Collapsed likelihood

The marginal likelihood is

$$p(v \mid b, \nu_0) = \int p(v \mid R, b) p(R \mid \nu_0) dR.$$

Using (4),

$$p(v \mid b, \nu_0) = \exp(Nb^\top v) \int p(v \mid R, b=0) \exp\left\{-\frac{N}{2}b^\top Rb\right\} p(R \mid \nu_0) dR. \quad (5)$$

Define the null marginal likelihood

$$p_0(v \mid \nu_0) = \int p(v \mid R, b=0) p(R \mid \nu_0) dR.$$

Then

$$p(v \mid b, \nu_0) = p_0(v \mid \nu_0) \exp(Nb^\top v) \mathbb{E}_{R \mid v, b=0, \nu_0} \left[\exp\left\{-\frac{N}{2}b^\top Rb\right\} \right]. \quad (6)$$

Under the null model,

$$v \mid R, b=0 \sim \mathcal{N}_p(0, R/N),$$

and by inverse-Wishart conjugacy,

$$R \mid v, b=0, \nu_0 \sim \text{IW}_p\left(\nu_0 \bar{R} + Nvv^\top, \nu_0 + p + 2\right).$$

Let

$$B_\nu = \nu_0 \bar{R} + Nvv^\top.$$

If

$$R \sim \text{IW}_p(B_\nu, \nu_0 + p + 2),$$

then for fixed b ,

$$b^\top Rb \stackrel{d}{=} \frac{b^\top B_\nu b}{\omega}, \quad \omega \sim \chi_{\nu_0+3}^2.$$

Hence the exact collapsed likelihood is

$$p(v \mid b, \nu_0) = p_0(v \mid \nu_0) \exp(Nb^\top v) \mathbb{E}_{\omega \sim \chi_{\nu_0+3}^2} \left[\exp\left\{-\frac{N}{2\omega}b^\top (\nu_0 \bar{R} + Nvv^\top)b\right\} \right]. \quad (7)$$

Define

$$Q_{\nu_0}(b) = b^\top (\nu_0 \bar{R} + Nvv^\top)b.$$

Then

$$p(v \mid b, \nu_0) = p_0(v \mid \nu_0) \exp(Nb^\top v) \mathbb{E}_{\omega \sim \chi_{\nu_0+3}^2} \left[\exp\left\{-\frac{NQ_{\nu_0}(b)}{2\omega}\right\} \right].$$

The null marginal likelihood is a multivariate t density:

$$p_0(v \mid \nu_0) = t_{\nu_0+2}\left(v; 0, \frac{\nu_0}{N(\nu_0+2)}\bar{R}\right).$$

Equivalently, if

$$r_0 = v^\top \bar{R}^{-1} v, \quad \Sigma_\nu = \frac{\nu_0}{N(\nu_0 + 2)} \bar{R},$$

then

$$\begin{aligned} \log p_0(v \mid \nu_0) &= \log \Gamma\left(\frac{\nu_0 + p + 2}{2}\right) - \log \Gamma\left(\frac{\nu_0 + 2}{2}\right) - \frac{p}{2} \log\{(\nu_0 + 2)\pi\} - \frac{1}{2} \log |\Sigma_\nu| \\ &\quad - \frac{\nu_0 + p + 2}{2} \log\left(1 + \frac{N}{\nu_0} r_0\right). \end{aligned} \quad (8)$$

3 Variational lower bound

For any variational density $q(\omega)$,

$$\log \mathbb{E}_{\omega \sim \chi_{\nu_0+3}^2} \left[\exp \left\{ -\frac{NQ_{\nu_0}(b)}{2\omega} \right\} \right] \geq -\frac{N}{2} \mathbb{E}_q(\omega^{-1}) Q_{\nu_0}(b) - \text{KL}\{q(\omega) \parallel \chi_{\nu_0+3}^2\}. \quad (9)$$

Therefore,

$$\log p(v \mid b, \nu_0) \geq \log p_0(v \mid \nu_0) + Nb^\top v - \frac{N}{2} \mathbb{E}_q(\omega^{-1}) Q_{\nu_0}(b) - \text{KL}\{q(\omega) \parallel \chi_{\nu_0+3}^2\}. \quad (10)$$

This bound becomes tight for the optimal density

$$q^*(\omega) \propto \chi_{\nu_0+3}^2(\omega) \exp \left\{ -\frac{NQ_{\nu_0}(b)}{2\omega} \right\}.$$

Since

$$\chi_d^2(\omega) \propto \omega^{d/2-1} \exp(-\omega/2),$$

we have

$$q^*(\omega) = \text{GIG} \left(\lambda = \frac{\nu_0 + 3}{2}, \chi = NQ_{\nu_0}(b), \psi = 1 \right),$$

where the GIG density is

$$\text{GIG}(\lambda, \chi, \psi) \propto \omega^{\lambda-1} \exp \left\{ -\frac{1}{2} \left(\frac{\chi}{\omega} + \psi\omega \right) \right\}.$$

4 Single-effect model

Now consider $L = 1$. Let

$$\gamma \in \{1, \dots, p\}, \quad b = \beta e_\gamma.$$

Assume

$$P(\gamma = j) = \pi_j, \quad \sum_{j=1}^p \pi_j = 1,$$

and

$$\beta \mid \gamma = j, \sigma_0^2 \sim \mathcal{N}(0, \sigma_0^2).$$

For $\gamma = j$,

$$b = \beta e_j,$$

so

$$b^\top v = \beta v_j.$$

Also,

$$Q_{\nu_0}(\beta e_j) = \beta^2 e_j^\top (\nu_0 \bar{R} + N v v^\top) e_j \quad (11)$$

$$= \beta^2 (\nu_0 \bar{R}_{jj} + N v_j^2). \quad (12)$$

Define

$$a_j(\nu_0) = \nu_0 \bar{R}_{jj} + N v_j^2.$$

If $\bar{R}$ is a correlation matrix, then typically $\bar{R}_{jj} = 1$, giving

$$a_j(\nu_0) = \nu_0 + N v_j^2.$$

The exact collapsed likelihood conditional on $(\gamma = j, \beta)$ is

$$p(v \mid \gamma = j, \beta, \nu_0) = p_0(v \mid \nu_0) \exp(N \beta v_j) \mathbb{E}_{\omega \sim \chi_{\nu_0+3}^2} \left[\exp \left\{ -\frac{N a_j(\nu_0) \beta^2}{2\omega} \right\} \right]. \quad (13)$$

The augmented joint density is

$$\begin{aligned} p(v, \gamma = j, \beta, \omega \mid \nu_0, \sigma_0^2) &= p_0(v \mid \nu_0) \pi_j \mathcal{N}(\beta; 0, \sigma_0^2) \chi_{\nu_0+3}^2(\omega) \\ &\quad \times \exp \left\{ N \beta v_j - \frac{N a_j(\nu_0) \beta^2}{2\omega} \right\}. \end{aligned} \quad (14)$$

5 Mean-field variational family

Use the variational family

$$q(\gamma = j, \beta, \omega) = \alpha_j q_j(\beta) q_j(\omega),$$

where

$$\alpha_j \geq 0, \quad \sum_{j=1}^p \alpha_j = 1.$$

This should be read as

$$q(\gamma = j) = \alpha_j, \quad q(\beta, \omega \mid \gamma = j) = q_j(\beta) q_j(\omega).$$

The factorization allows the variational distributions of β and ω to depend on the candidate causal SNP j . This is important because each j has a different v_j and $a_j(\nu_0)$.

Take

$$q_j(\beta) = \mathcal{N}(\mu_j, s_j^2),$$

and

$$q_j(\omega) = \text{GIG}(\lambda_j, \chi_j, \psi_j).$$

In the coordinate-optimal update below,

$$\lambda_j = \frac{\nu_0 + 3}{2}, \quad \psi_j = 1.$$

Define

$$m_{2j} = \mathbb{E}_{q_j}(\beta^2) = \mu_j^2 + s_j^2,$$

and

$$\eta_j = \mathbb{E}_{q_j}(\omega^{-1}).$$

The ELBO is

$$\mathcal{L}(\nu_0, \sigma_0^2, q) = \log p_0(v \mid \nu_0) + \sum_{j=1}^p \alpha_j [\log \pi_j - \log \alpha_j + \mathcal{F}_j], \quad (15)$$

where

$$\begin{aligned} \mathcal{F}_j &= \mathbb{E}_{q_j(\beta)} [\log \mathcal{N}(\beta; 0, \sigma_0^2) - \log q_j(\beta)] + N v_j \mathbb{E}_{q_j}(\beta) \\ &\quad - \frac{N}{2} a_j(\nu_0) \mathbb{E}_{q_j}(\beta^2) \mathbb{E}_{q_j}(\omega^{-1}) - \text{KL}\{q_j(\omega) \parallel \chi_{\nu_0+3}^2\}. \end{aligned} \quad (16)$$

Since $q_j(\beta) = \mathcal{N}(\mu_j, s_j^2)$,

$$\mathbb{E}_{q_j(\beta)} [\log \mathcal{N}(\beta; 0, \sigma_0^2) - \log q_j(\beta)] = \frac{1}{2} \left[\log \left(\frac{s_j^2}{\sigma_0^2} \right) + 1 - \frac{m_{2j}}{\sigma_0^2} \right]. \quad (17)$$

Therefore,

$$\mathcal{F}_j = \frac{1}{2} \left[\log \left(\frac{s_j^2}{\sigma_0^2} \right) + 1 - \frac{m_{2j}}{\sigma_0^2} \right] + N v_j \mu_j - \frac{N}{2} a_j(\nu_0) m_{2j} \eta_j - \text{KL}\{q_j(\omega) \parallel \chi_{\nu_0+3}^2\}. \quad (18)$$

6 Coordinate updates for fixed ν_0 and σ_0^2

For fixed ν_0 , σ_0^2 , and $q_j(\omega)$, the optimal $q_j(\beta)$ satisfies

$$q_j(\beta) \propto \mathcal{N}(\beta; 0, \sigma_0^2) \exp \left\{ N v_j \beta - \frac{N}{2} a_j(\nu_0) \eta_j \beta^2 \right\}.$$

Thus

$$q_j(\beta) = \mathcal{N}(\mu_j, s_j^2),$$

where

$$s_j^2 = \left[\frac{1}{\sigma_0^2} + N a_j(\nu_0) \eta_j \right]^{-1}, \quad (19)$$

$$\mu_j = s_j^2 N v_j. \quad (20)$$

Then

$$m_{2j} = \mu_j^2 + s_j^2.$$

For fixed $q_j(\beta)$, the optimal $q_j(\omega)$ is

$$q_j(\omega) \propto \chi_{\nu_0+3}^2(\omega) \exp \left\{ -\frac{N a_j(\nu_0) m_{2j}}{2\omega} \right\}. \quad (21)$$

Therefore,

$$q_j(\omega) = \text{GIG}(\lambda_j, \chi_j, \psi_j),$$

with

$$\lambda_j = \frac{\nu_0 + 3}{2}, \quad (22)$$

$$\chi_j = Na_j(\nu_0)m_{2j}, \quad (23)$$

$$\psi_j = 1. \quad (24)$$

For this GIG distribution,

$$\eta_j = \mathbb{E}_{q_j}(\omega^{-1}) = \left(\frac{\psi_j}{\chi_j}\right)^{1/2} \frac{K_{\lambda_j-1}(\sqrt{\chi_j\psi_j})}{K_{\lambda_j}(\sqrt{\chi_j\psi_j})}.$$

Since $\psi_j = 1$,

$$\eta_j = \frac{1}{\sqrt{\chi_j}} \frac{K_{(\nu_0+1)/2}(\sqrt{\chi_j})}{K_{(\nu_0+3)/2}(\sqrt{\chi_j})}.$$

If $\chi_j = 0$, use the limit

$$\eta_j = \frac{1}{\nu_0 + 1}.$$

For fixed local factors, the optimal posterior inclusion weights are

$$\alpha_j = \frac{\pi_j \exp(\mathcal{F}_j)}{\sum_{k=1}^p \pi_k \exp(\mathcal{F}_k)}.$$

The variational posterior for b is therefore

$$q(b) = \sum_{j=1}^p \alpha_j \int \delta_{\beta e_j}(b) q_j(\beta) d\beta.$$

Consequently,

$$\mathbb{E}_q(b_j) = \alpha_j \mu_j,$$

and

$$\mathbb{E}_q(b_j^2) = \alpha_j m_{2j}.$$

7 KL term for $q_j(\omega)$

Let

$$d = \nu_0 + 3, \quad \lambda = \frac{d}{2} = \frac{\nu_0 + 3}{2}.$$

The chi-square density is

$$\chi_d^2(\omega) = \frac{1}{2^\lambda \Gamma(\lambda)} \omega^{\lambda-1} \exp(-\omega/2).$$

The coordinate-optimal variational density is

$$q_j(\omega) = \text{GIG}(\lambda, \chi_j, 1).$$

Its density is

$$q_j(\omega) = \frac{\chi_j^{-\lambda/2}}{2K_\lambda(\sqrt{\chi_j})} \omega^{\lambda-1} \exp\left\{-\frac{1}{2}\left(\frac{\chi_j}{\omega} + \omega\right)\right\}.$$

Therefore,

$$\text{KL}\{q_j(\omega) \parallel \chi_d^2\} = \log \Gamma(\lambda) + (\lambda - 1) \log 2 - \frac{\lambda}{2} \log \chi_j - \log K_\lambda(\sqrt{\chi_j}) - \frac{\chi_j}{2} \eta_j. \quad (25)$$

When $\chi_j \rightarrow 0$, this KL goes to zero, as expected.

8 Empirical-Bayes updates for ν_0 and σ_0^2

We treat ν_0 and σ_0^2 as empirical-Bayes hyperparameters.

The variational empirical-Bayes objective is

$$\widehat{\ell}_{\text{VEB}}(\nu_0, \sigma_0^2) = \max_q \mathcal{L}(\nu_0, \sigma_0^2, q).$$

For fixed ν_0 , the update for σ_0^2 has a closed form. Since

$$\sum_{j=1}^p \alpha_j = 1,$$

the part of the ELBO depending on σ_0^2 is

$$-\frac{1}{2} \sum_{j=1}^p \alpha_j \left[\log \sigma_0^2 + \frac{m_{2j}}{\sigma_0^2} \right].$$

Maximizing with respect to σ_0^2 gives

$$\sigma_0^2 \leftarrow \sum_{j=1}^p \alpha_j m_{2j}.$$

There is generally no closed-form update for ν_0 , because ν_0 appears in

$$\log p_0(v \mid \nu_0),$$

$$a_j(\nu_0) = \nu_0 \bar{R}_{jj} + N v_j^2,$$

$$\lambda_j = \frac{\nu_0 + 3}{2},$$

and

$$\text{KL}\{q_j(\omega) \parallel \chi_{\nu_0+3}^2\}.$$

Thus update ν_0 numerically:

$$\nu_0 \leftarrow \arg \max_{\nu > 0} \mathcal{L}(\nu, \sigma_0^2, q).$$

In practice, it is often more stable to optimize over

$$\rho = \log \nu_0,$$

so that

$$\nu_0 = \exp(\rho).$$

Alternatively, use a one-dimensional grid:

$$\nu_0 \in \{\nu^{(1)}, \dots, \nu^{(G)}\},$$

and choose the grid point with the largest optimized ELBO.

9 VEB algorithm for $L = 1$

1. **Input:** summary statistic v , sample size N , reference LD matrix $\bar{R}$, prior weights π_j , initial values $\nu_0^{(0)}$ and $\sigma_0^{2(0)}$.
2. **Initialize:** for each j , initialize

$$\alpha_j^{(0)} = \pi_j, \quad \eta_j^{(0)} = \frac{1}{\nu_0^{(0)} + 1}.$$

3. **Repeat until convergence:**

- (a) For each SNP $j = 1, \dots, p$, compute

$$a_j(\nu_0) = \nu_0 \bar{R}_{jj} + N v_j^2.$$

- (b) Update $q_j(\beta)$:

$$\begin{aligned} s_j^2 &= \left[\frac{1}{\sigma_0^2} + N a_j(\nu_0) \eta_j \right]^{-1}, \\ \mu_j &= s_j^2 N v_j, \\ m_{2j} &= \mu_j^2 + s_j^2. \end{aligned}$$

- (c) Update $q_j(\omega)$:

$$\lambda_j = \frac{\nu_0 + 3}{2}, \quad \chi_j = N a_j(\nu_0) m_{2j}, \quad \psi_j = 1.$$

Set

$$q_j(\omega) = \text{GIG}(\lambda_j, \chi_j, 1).$$

- (d) Update

$$\eta_j = \mathbb{E}_{q_j}(\omega^{-1}) = \frac{1}{\sqrt{\chi_j}} \frac{K_{(\nu_0+1)/2}(\sqrt{\chi_j})}{K_{(\nu_0+3)/2}(\sqrt{\chi_j})}.$$

If χ_j is numerically close to zero, use

$$\eta_j = \frac{1}{\nu_0 + 1}.$$

- (e) Compute the local objective

$$\mathcal{F}_j = \frac{1}{2} \left[\log \left(\frac{s_j^2}{\sigma_0^2} \right) + 1 - \frac{m_{2j}}{\sigma_0^2} \right] + N v_j \mu_j - \frac{N}{2} a_j(\nu_0) m_{2j} \eta_j - \text{KL}\{q_j(\omega) \parallel \chi_{\nu_0+3}^2\}.$$

- (f) Update posterior inclusion probabilities:

$$\alpha_j = \frac{\pi_j \exp(\mathcal{F}_j)}{\sum_{k=1}^p \pi_k \exp(\mathcal{F}_k)}.$$

- (g) Update the prior variance:

$$\sigma_0^2 \leftarrow \sum_{j=1}^p \alpha_j m_{2j}.$$

(h) Update the LD-mismatch parameter:

$$\nu_0 \leftarrow \arg \max_{\nu > 0} \mathcal{L}(\nu, \sigma_0^2, q).$$

This step is one-dimensional and can be done by grid search or numerical optimization over $\log \nu$.

4. Output:

$$\widehat{\nu}_0, \quad \widehat{\sigma}_0^2, \quad \alpha_j, \quad q_j(\beta) = \mathcal{N}(\mu_j, s_j^2), \quad q_j(\omega) = \text{GIG}(\lambda_j, \chi_j, 1).$$

The posterior mean of the effect vector is

$$\mathbb{E}_q(b_j) = \alpha_j \mu_j.$$

The posterior second moment is

$$\mathbb{E}_q(b_j^2) = \alpha_j(\mu_j^2 + s_j^2).$$

10 Variant with a shared $q(\omega)$

A simpler approximation uses a single variational distribution for ω , shared across all candidate causal SNPs:

$$q(\gamma = j, \beta, \omega) = \alpha_j q_j(\beta) q(\omega).$$

Write

$$q(\omega) = \text{GIG}(\lambda, \chi, \psi), \quad \eta = \mathbb{E}_q(\omega^{-1}).$$

The ELBO becomes

$$\begin{aligned} \mathcal{L}_{\text{shared}}(\nu_0, \sigma_0^2, q) &= \log p_0(v \mid \nu_0) - \text{KL}\{q(\omega) \parallel \chi_{\nu_0+3}^2\} \\ &\quad + \sum_{j=1}^p \alpha_j [\log \pi_j - \log \alpha_j + \mathcal{G}_j], \end{aligned} \tag{26}$$

where

$$\mathcal{G}_j = \frac{1}{2} \left[\log \left(\frac{s_j^2}{\sigma_0^2} \right) + 1 - \frac{m_{2j}}{\sigma_0^2} \right] + N v_j \mu_j - \frac{N}{2} a_j(\nu_0) m_{2j} \eta. \tag{27}$$

The only difference from (18) is that the same η is used for every j , and the KL penalty for ω is global rather than local.

For fixed $q(\omega)$, the update for $q_j(\beta)$ is

$$s_j^2 = \left[\frac{1}{\sigma_0^2} + N a_j(\nu_0) \eta \right]^{-1}, \tag{28}$$

$$\mu_j = s_j^2 N v_j, \tag{29}$$

$$m_{2j} = \mu_j^2 + s_j^2. \tag{30}$$

For fixed $\{\alpha_j, q_j(\beta)\}_{j=1}^p$, the coordinate-optimal shared $q(\omega)$ is

$$q(\omega) \propto \chi_{\nu_0+3}^2(\omega) \exp \left\{ -\frac{N}{2\omega} \sum_{j=1}^p \alpha_j a_j(\nu_0) m_{2j} \right\}.$$

Thus

$$q(\omega) = \text{GIG}(\lambda, \chi, 1),$$

with

$$\lambda = \frac{\nu_0 + 3}{2}, \quad (31)$$

$$\chi = N \sum_{j=1}^p \alpha_j a_j(\nu_0) m_{2j}. \quad (32)$$

The shared inverse moment is

$$\eta = \frac{1}{\sqrt{\chi}} \frac{K_{(\nu_0+1)/2}(\sqrt{\chi})}{K_{(\nu_0+3)/2}(\sqrt{\chi})}.$$

If χ is numerically close to zero, use

$$\eta = \frac{1}{\nu_0 + 1}.$$

The posterior inclusion probabilities are updated using

$$\alpha_j = \frac{\pi_j \exp(\mathcal{G}_j)}{\sum_{k=1}^p \pi_k \exp(\mathcal{G}_k)}.$$

Shared- ω VEB algorithm for $L = 1$

1. **Input:** summary statistic v , sample size N , reference LD matrix $\bar{R}$, prior weights π_j , initial values $\nu_0^{(0)}$ and $\sigma_0^{2(0)}$.

2. **Initialize:**

$$\alpha_j^{(0)} = \pi_j, \quad \eta^{(0)} = \frac{1}{\nu_0^{(0)} + 1}.$$

3. **Repeat until convergence:**

- (a) For each SNP $j = 1, \dots, p$, compute

$$a_j(\nu_0) = \nu_0 \bar{R}_{jj} + N v_j^2.$$

- (b) Update $q_j(\beta)$ using the shared η :

$$s_j^2 = \left[\frac{1}{\sigma_0^2} + N a_j(\nu_0) \eta \right]^{-1}, \quad \mu_j = s_j^2 N v_j, \quad m_{2j} = \mu_j^2 + s_j^2.$$

- (c) Update the posterior inclusion probabilities:

$$\mathcal{G}_j = \frac{1}{2} \left[\log \left(\frac{s_j^2}{\sigma_0^2} \right) + 1 - \frac{m_{2j}}{\sigma_0^2} \right] + N v_j \mu_j - \frac{N}{2} a_j(\nu_0) m_{2j} \eta,$$

$$\alpha_j = \frac{\pi_j \exp(\mathcal{G}_j)}{\sum_{k=1}^p \pi_k \exp(\mathcal{G}_k)}.$$

(d) Update the shared $q(\omega)$:

$$\lambda = \frac{\nu_0 + 3}{2}, \quad \chi = N \sum_{j=1}^p \alpha_j a_j(\nu_0) m_{2j}, \quad q(\omega) = \text{GIG}(\lambda, \chi, 1).$$

(e) Update the shared inverse moment:

$$\eta = \frac{1}{\sqrt{\chi}} \frac{K_{(\nu_0+1)/2}(\sqrt{\chi})}{K_{(\nu_0+3)/2}(\sqrt{\chi})}.$$

If χ is numerically close to zero, use

$$\eta = \frac{1}{\nu_0 + 1}.$$

(f) Update the prior variance:

$$\sigma_0^2 \leftarrow \sum_{j=1}^p \alpha_j m_{2j}.$$

(g) Update the LD-mismatch parameter:

$$\nu_0 \leftarrow \arg \max_{\nu > 0} \mathcal{L}_{\text{shared}}(\nu, \sigma_0^2, q).$$

4. Output:

$$\hat{\nu}_0, \quad \widehat{\sigma}_0^2, \quad \alpha_j, \quad q_j(\beta) = \mathcal{N}(\mu_j, s_j^2), \quad q(\omega) = \text{GIG}(\lambda, \chi, 1).$$

11 Variant replacing ω by $\nu_0 + 3$

An even simpler approximation replaces the random variable

$$\omega \sim \chi_{\nu_0+3}^2$$

by its mean,

$$\mathbb{E}(\omega) = \nu_0 + 3.$$

Equivalently, replace every occurrence of ω^{-1} in the quadratic penalty by

$$\frac{1}{\nu_0 + 3}.$$

For $\gamma = j$, this gives the plug-in approximation

$$p(v \mid \gamma = j, \beta, \nu_0) \approx p_0(v \mid \nu_0) \exp \left\{ N \beta v_j - \frac{N a_j(\nu_0) \beta^2}{2(\nu_0 + 3)} \right\}.$$

There is no variational distribution for ω and no KL penalty for ω .

Use the variational family

$$q(\gamma = j, \beta) = \alpha_j q_j(\beta), \quad q_j(\beta) = \mathcal{N}(\mu_j, s_j^2).$$

The plug-in objective is

$$\mathcal{L}_{\text{plug}}(\nu_0, \sigma_0^2, q) = \log p_0(v \mid \nu_0) + \sum_{j=1}^p \alpha_j [\log \pi_j - \log \alpha_j + \mathcal{H}_j], \quad (33)$$

where

$$\mathcal{H}_j = \frac{1}{2} \left[\log \left(\frac{s_j^2}{\sigma_0^2} \right) + 1 - \frac{m_{2j}}{\sigma_0^2} \right] + N v_j \mu_j - \frac{N a_j(\nu_0) m_{2j}}{2(\nu_0 + 3)}. \quad (34)$$

For fixed ν_0 and σ_0^2 , the coordinate update for $q_j(\beta)$ is

$$s_j^2 = \left[\frac{1}{\sigma_0^2} + \frac{N a_j(\nu_0)}{\nu_0 + 3} \right]^{-1}, \quad (35)$$

$$\mu_j = s_j^2 N v_j, \quad (36)$$

$$m_{2j} = \mu_j^2 + s_j^2. \quad (37)$$

The posterior inclusion probabilities are

$$\alpha_j = \frac{\pi_j \exp(\mathcal{H}_j)}{\sum_{k=1}^p \pi_k \exp(\mathcal{H}_k)}.$$

Plug-in- ω VEB algorithm for $L = 1$

1. **Input:** summary statistic v , sample size N , reference LD matrix $\bar{R}$, prior weights π_j , initial values $\nu_0^{(0)}$ and $\sigma_0^{2(0)}$.

2. **Initialize:**

$$\alpha_j^{(0)} = \pi_j.$$

3. **Repeat until convergence:**

(a) For each SNP $j = 1, \dots, p$, compute

$$a_j(\nu_0) = \nu_0 \bar{R}_{jj} + N v_j^2.$$

(b) Update $q_j(\beta)$:

$$s_j^2 = \left[\frac{1}{\sigma_0^2} + \frac{N a_j(\nu_0)}{\nu_0 + 3} \right]^{-1}, \quad \mu_j = s_j^2 N v_j, \quad m_{2j} = \mu_j^2 + s_j^2.$$

(c) Compute the plug-in local objective:

$$\mathcal{H}_j = \frac{1}{2} \left[\log \left(\frac{s_j^2}{\sigma_0^2} \right) + 1 - \frac{m_{2j}}{\sigma_0^2} \right] + N v_j \mu_j - \frac{N a_j(\nu_0) m_{2j}}{2(\nu_0 + 3)}.$$

(d) Update posterior inclusion probabilities:

$$\alpha_j = \frac{\pi_j \exp(\mathcal{H}_j)}{\sum_{k=1}^p \pi_k \exp(\mathcal{H}_k)}.$$

(e) Update the prior variance:

$$\sigma_0^2 \leftarrow \sum_{j=1}^p \alpha_j m_{2j}.$$

(f) Update the LD-mismatch parameter:

$$\nu_0 \leftarrow \arg \max_{\nu > 0} \mathcal{L}_{\text{plug}}(\nu, \sigma_0^2, q).$$

4. **Output:**

$$\widehat{\nu}_0, \quad \widehat{\sigma}_0^2, \quad \alpha_j, \quad q_j(\beta) = \mathcal{N}(\mu_j, s_j^2).$$

12 SuSiE version with a shared $q(\omega)$

Now consider a multi-effect model with

$$b = \sum_{\ell=1}^L b_\ell, \quad b_\ell = \beta_\ell e_{\gamma_\ell}.$$

For effect ℓ , let

$$P(\gamma_\ell = j) = \pi_{\ell j}, \quad \beta_\ell \mid \gamma_\ell = j, \sigma_\ell^2 \sim \mathcal{N}(0, \sigma_\ell^2).$$

Write

$$A_\nu = \nu_0 \bar{R} + N v v^\top, \quad Q_\nu(b) = b^\top A_\nu b.$$

The collapsed lower bound with a single shared $q(\omega)$ is

$$\log p(v \mid b, \nu_0) \geq \log p_0(v \mid \nu_0) + N b^\top v - \frac{N}{2} \eta b^\top A_\nu b - \text{KL}\{q(\omega) \parallel \chi_{\nu_0+3}^2\}, \quad (38)$$

where

$$\eta = \mathbb{E}_q(\omega^{-1}).$$

Use the mean-field family

$$q(b, \omega) = q(\omega) \prod_{\ell=1}^L q_\ell(\gamma_\ell, \beta_\ell),$$

with

$$q_\ell(\gamma_\ell = j, \beta_\ell) = \alpha_{\ell j} q_{\ell j}(\beta_\ell), \quad q_{\ell j}(\beta_\ell) = \mathcal{N}(\mu_{\ell j}, s_{\ell j}^2).$$

Define

$$m_{\ell j} = \alpha_{\ell j} \mu_{\ell j}, \quad m_\ell = (m_{\ell 1}, \dots, m_{\ell p})^\top, \quad m = \sum_{\ell=1}^L m_\ell,$$

and

$$m_{2, \ell j} = \mathbb{E}_{q_{\ell j}}(\beta_\ell^2) = \mu_{\ell j}^2 + s_{\ell j}^2.$$

The shared- ω SuSiE ELBO can be written as

$$\begin{aligned}\mathcal{L}_{\text{SuSiE}}(\nu_0, \{\sigma_\ell^2\}, q) &= \log p_0(v \mid \nu_0) - \text{KL}\{q(\omega) \parallel \chi_{\nu_0+3}^2\} \\ &\quad + N v^\top m - \frac{N}{2} \eta S_Q \\ &\quad + \sum_{\ell=1}^L \sum_{j=1}^p \alpha_{\ell j} [\log \pi_{\ell j} - \log \alpha_{\ell j} \\ &\quad \quad + \frac{1}{2} \left\{ \log \left(\frac{s_{\ell j}^2}{\sigma_\ell^2} \right) + 1 - \frac{m_{2,\ell j}}{\sigma_\ell^2} \right\}] ,\end{aligned}\tag{39}$$

where

$$S_Q = \mathbb{E}_q\{b^\top A_\nu b\}.$$

An efficient expression for this expectation is

$$S_Q = m^\top A_\nu m + \sum_{\ell=1}^L \left[\sum_{j=1}^p \alpha_{\ell j} m_{2,\ell j} (A_\nu)_{jj} - m_\ell^\top A_\nu m_\ell \right].$$

The second term adds back the within-effect posterior variance and removes the squared posterior mean already counted in $m^\top A_\nu m$.

Coordinate update for one effect

When updating effect ℓ , hold the other effects and $q(\omega)$ fixed. Let

$$m_{-\ell} = \sum_{k \neq \ell} m_k = m - m_\ell, \quad c_\ell = A_\nu m_{-\ell}.$$

For candidate SNP j , the terms involving β_ℓ are

$$N \beta_\ell v_j - N \eta \beta_\ell (c_\ell)_j - \frac{N}{2} \eta (A_\nu)_{jj} \beta_\ell^2 + \log \mathcal{N}(\beta_\ell; 0, \sigma_\ell^2).$$

Thus

$$q_{\ell j}(\beta_\ell) = \mathcal{N}(\mu_{\ell j}, s_{\ell j}^2),$$

where

$$s_{\ell j}^2 = \left[\frac{1}{\sigma_\ell^2} + N \eta (A_\nu)_{jj} \right]^{-1}, \tag{40}$$

$$\mu_{\ell j} = s_{\ell j}^2 N [v_j - \eta (c_\ell)_j], \tag{41}$$

$$m_{2,\ell j} = \mu_{\ell j}^2 + s_{\ell j}^2. \tag{42}$$

The coordinate score for $\gamma_\ell = j$ is

$$\begin{aligned}\mathcal{D}_{\ell j} &= \frac{1}{2} \left[\log \left(\frac{s_{\ell j}^2}{\sigma_\ell^2} \right) + 1 - \frac{m_{2,\ell j}}{\sigma_\ell^2} \right] \\ &\quad + N \mu_{\ell j} [v_j - \eta (c_\ell)_j] - \frac{N}{2} \eta (A_\nu)_{jj} m_{2,\ell j}.\end{aligned}\tag{43}$$

Therefore,

$$\alpha_{\ell j} = \frac{\pi_{\ell j} \exp(\mathcal{D}_{\ell j})}{\sum_{k=1}^p \pi_{\ell k} \exp(\mathcal{D}_{\ell k})}.$$

Shared $q(\omega)$ update

For fixed effect factors, the optimal shared $q(\omega)$ is

$$q(\omega) \propto \chi_{\nu_0+3}^2(\omega) \exp \left\{ -\frac{NS_Q}{2\omega} \right\}.$$

Thus

$$q(\omega) = \text{GIG}(\lambda, \chi, 1),$$

with

$$\lambda = \frac{\nu_0 + 3}{2}, \quad (44)$$

$$\chi = NS_Q. \quad (45)$$

The shared inverse moment is

$$\eta = \frac{1}{\sqrt{\chi}} \frac{K_{(\nu_0+1)/2}(\sqrt{\chi})}{K_{(\nu_0+3)/2}(\sqrt{\chi})}.$$

If χ is numerically close to zero, use

$$\eta = \frac{1}{\nu_0 + 1}.$$

Empirical-Bayes updates

For fixed ν_0 , a natural update for the effect-specific prior variance is

$$\sigma_\ell^2 \leftarrow \sum_{j=1}^p \alpha_{\ell j} m_{2,\ell j}.$$

There is generally no closed-form update for ν_0 in this shared- ω collapsed objective. A practical update is

$$\nu_0 \leftarrow \arg \max_{\nu > 0} \mathcal{L}_{\text{SuSiE}}(\nu, \{\sigma_\ell^2\}, q),$$

or a grid search over candidate values of ν . When ν_0 changes, A_ν , λ , χ , η , and the null term $\log p_0(v \mid \nu)$ should be recomputed.

Shared- ω collapsed SuSiE algorithm

1. **Input:** summary statistic v , sample size N , reference LD matrix $\bar{R}$, number of effects L , prior weights $\pi_{\ell j}$, initial values $\nu_0^{(0)}$ and $(\sigma_\ell^2)^{(0)}$.
2. **Initialize:** for each effect ℓ , initialize

$$\alpha_{\ell j}^{(0)} = \pi_{\ell j}, \quad \mu_{\ell j}^{(0)} = 0, \quad s_{\ell j}^{2(0)} = (\sigma_\ell^2)^{(0)}.$$

Set

$$\eta^{(0)} = \frac{1}{\nu_0^{(0)} + 1}.$$

3. **Repeat until convergence:**

(a) Form

$$A_\nu = \nu_0 \bar{R} + N v v^\top.$$

(b) For each effect $\ell = 1, \dots, L$, perform an IBSS update:

i. Compute

$$m_\ell = (\alpha_{\ell 1} \mu_{\ell 1}, \dots, \alpha_{\ell p} \mu_{\ell p})^\top, \quad m_{-\ell} = \sum_{k \neq \ell} m_k, \quad c_\ell = A_\nu m_{-\ell}.$$

ii. For each SNP $j = 1, \dots, p$, update

$$\begin{aligned} s_{\ell j}^2 &= \left[\frac{1}{\sigma_\ell^2} + N \eta(A_\nu)_{jj} \right]^{-1}, \\ \mu_{\ell j} &= s_{\ell j}^2 N [v_j - \eta(c_\ell)_j], \\ m_{2, \ell j} &= \mu_{\ell j}^2 + s_{\ell j}^2. \end{aligned}$$

iii. Compute $\mathcal{D}_{\ell j}$ and update

$$\alpha_{\ell j} = \frac{\pi_{\ell j} \exp(\mathcal{D}_{\ell j})}{\sum_{k=1}^p \pi_{\ell k} \exp(\mathcal{D}_{\ell k})}.$$

(c) Recompute

$$m = \sum_{\ell=1}^L m_\ell$$

and

$$S_Q = m^\top A_\nu m + \sum_{\ell=1}^L \left[\sum_{j=1}^p \alpha_{\ell j} m_{2, \ell j} (A_\nu)_{jj} - m_\ell^\top A_\nu m_\ell \right].$$

(d) Update the shared $q(\omega)$:

$$\lambda = \frac{\nu_0 + 3}{2}, \quad \chi = N S_Q, \quad q(\omega) = \text{GIG}(\lambda, \chi, 1).$$

Then update

$$\eta = \mathbb{E}_q(\omega^{-1}).$$

(e) Optionally update

$$\sigma_\ell^2 \leftarrow \sum_{j=1}^p \alpha_{\ell j} m_{2, \ell j}, \quad \ell = 1, \dots, L.$$

(f) Optionally update ν_0 by grid search or one-dimensional optimization of $\mathcal{L}_{\text{SuSiE}}$.

4. Output:

$$\hat{\nu}_0, \quad \hat{\sigma}_1^2, \dots, \hat{\sigma}_L^2, \quad \alpha_{\ell j}, \quad q_{\ell j}(\beta_\ell) = \mathcal{N}(\mu_{\ell j}, s_{\ell j}^2), \quad q(\omega) = \text{GIG}(\lambda, \chi, 1).$$

The posterior inclusion probability for SNP j is

$$\text{PIP}_j = 1 - \prod_{\ell=1}^L (1 - \alpha_{\ell j}).$$

13 Comments

For $L = 1$, exact one-dimensional integration over β is also feasible. The VEB approximation is still useful because it provides a computational template that generalizes naturally to multi-effect SuSiE-style models.

The empirical-Bayes estimate $\hat{\nu}_0$ should be interpreted as a working LD-mismatch parameter. Smaller values of ν_0 allow more uncertainty around the reference LD matrix $\bar{R}$, whereas larger values concentrate R more tightly around $\bar{R}$.